25900587 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEPSY200 - BIOPSYCHOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Question- Choose the most appropriate answer

Answer all questions

Each question carries **1** marks

1. Which of the following provides the most detailed three-dimensional view of the structure of the living human brain? [K] / [CO1]

   i. PET ii . angiography

   iii. EEG iv. MRI

2. The midbrain is a part of which larger brain division? [K] / [CO2]

   i. Cerebellum ii . Diencephalon

   iii. Brainstem iv. Cerebrum

3. During maximum muscle contraction, EMG shows: [U] / [CO1]

   i. Dense interference pattern ii . Flat baseline

   iii. Alpha rhythm iv. Sleep spindles

4. Recurrent binge-eating without compensatory behaviour is characteristic of: [C] / [CO4]

   i. Rumination disorder ii . Binge-eating disorder

   iii. Pica iv. Bulimia Nervosa

5. Which process is responsible for the removal of neurotransmitters from the synaptic cleft by nearby glial cells? [U] / [CO2]

i. Diffusion ii . Exocytosis

iii. Reuptake iv. Enzymatic degradation

6. Which of the following is an example of a violation of ethical research practice in biopsychology? [K] / [CO1]

i.Using random sampling

ii.Reporting all results honestly

iii.Fabricating data to support a hypothesis

iv.Maintaining confidentiality

7. Damage to the cerebellum is most likely to cause [U] / [CO2]

i. Loss of memory ii . Difficulty in balance and coordination

iii. Impaired vision iv. Loss of smell

8. What is the name of the theory that proposes that testosterone is converted to estradiol in the brain to facilitate sexual differentiation? [An] / [CO4]

i. Aromatization Hypothesis ii . Hormone Conversion Theory

iii. Sexual Differentiation Syndrome iv. Androgen Insensitivity Syndrome

9. What does electrical stimulation of the brain typically cause? [U] / [CO1]

i. Permanent brain damage ii . Temporary activation of neurons

iii. Complete memory loss iv. Immediate unconsciousness

10. Which hypothalamic area is mainly responsible for triggering thrist? [K] / [CO5]

i. Lateral Hypothalamus ii . Supraoptic Nucleus

iii. Anterior Hypothalamus iv. Paraventricular Nucleus

11. Electrostatic pressure refers to: [K] / [CO2]

i. The force of water molecules moving ions

ii.The attraction and repulsion among charged ions

iii. The pressure of the cell membrane on ions

iv. The force of ATP hydrolysis

12. The HPA axis primarily regulates: [A] / [CO3]

i.Calcium homeostasis ii .Stress responses

iii.Milk secretion iv.Reproductive cycles

13. The primary hormones secreted by the adrenal medulla during stress are: [U] / [CO3]

i.Cortisol and aldosterone ii. Epinephrine and norepinephrine

iii. CRH and ACTH iv. Serotonin and dopamine

14. A postsynaptic potential that makes the postsynaptic neuron less likely to fire an action potential. [K] / [CO2]

i. Excitatory postsynaptic potential (EPSP)
ii . Synaptic potential
iii. Resting membrane potential
iv. Inhibitory postsynaptic potential(IPSP)

15. Select the metabolic condition most strongly linked with insulin resistance and memory impairment. [U] / [CO3]

i. Obesity
ii . Stress disorder
iii. Hyperthyroidism
iv. Diabetes

16. The nervous system responsible for voluntary control of skeletal muscles is [An] / [CO2]

i.Autonomic nervous system
ii .Somatic nervous system
iii.Enteric nervous system
iv.Parasympathetic system

17. What is the Papez circuit related to? [An] / [CO2]

i. Reflex actions
ii . Emotional expression and memory
iii. Vision pathways
iv. Motor learning

18. Insulin, a hormone that regulates blood glucose levels, is secreted by which gland? [K] / [CO3]

i. Thyroxine (T4)
ii . Cortisol
iii. Growth Hormone
iv. Adrenaline

19. Recognize the process of neurotransmitter release [An] / [CO2]

i.Exocytosis
ii . Enzymatic degradation
iii. Reuptake
iv. Coexistence

20. Positive feedback is beneficial in: [A] / [CO3]

i.Stress regulation
ii .Childbirth and lactation
iii.Long-term energy balance
iv.Temperature regulation

[1x20 = 20]

Part B

Short Answer type questions

Answer any **10** questions

Each question carries **3** marks

21. What is a reflex arc? Name its main components. [An] / [CO2]

22. Describe neuropathology and explain its role in understanding brain disorders. [U] / [CO1]

23. Explain some potential benefits of endorphin release. [U] / [CO3]

24. What is the significance of the limbic system in behaviour? [An] / [CO5]

25. What is the primary role of cortisol in the stress response? [K] / [CO3]

26. Define hemispheres. Mention the functions of the left and right hemisphere. [An] / [CO2]

27. Describe the role of acetylcholine. [An] / [CO2]

28. Describe how growth hormone affects learning. [U] / [CO3]

29. Explain satiation and satiety centre. [An] / [CO4]

30. Write a short note on neurotransmitters used in sympathetic and parasympathetic systems. [An] / [CO2]

31. Explain the role of the amygdala in processing sexually arousing stimuli. [An] / [CO4]

32. Describe how the influx of sodium (Na^+) and efflux of potassium (K^+) ions work together during the action potential. [An] / [CO2]

33. Discuss the role of Broca's area in speech and word formation. [K] / [CO2]

[3x10 = 30]

Part C

Essay type questions

Answer any **2** questions

Each question carries **10** marks

34. Explain how hormones influence sexual behaviour. [U] / [CO3]

35. Describe the structure of the forebrain and explain how each of its major components contributes to higher cognitive and physiological functions. [K] / [CO2]

36. Structural Differences between the Male and Female Hypothalamus. [An] / [CO4]

37. Illustrate the structure of synapse in detail and list out the types of synapses. [An] / [CO2]

[10x2 = 20]